

Dry Cannabis Flower for Medical use·*Cannabis sativa* L.·Ph. Eur. Monograph 3028·*Cannabis flos*

THC-dominant

Apples & Bananas

THC 15 : CBD 1

Grade IV

Lab Sample ID

Лаб. ИД на примерок

PP-QCL-26-042

Production Batch №

Производна серија №

P060052

Requested Test

Побарано испитување

Assay — Cannabinoid Content (Params 4–6)

Specification Ref.

Референца на спецификација

QCSP-IMB-001 v.02 · §4–6

01 Analytical Method | Аналитички метод

Method Метода	Method validation status: VAL · ICH Q2 Dossier MV-PP-HPLC-001
Assay — Total Cannabinoid Content by HPLC(Ph. Eur. 2.2.29 · mon. 3028)	
Sample preparation: 0.200 g homogenised, powdered drug (355 µm) extracted with 40.0 mL methanol R (30 min shaking + 15 min sonication), made to 50.0 mL; centrifuged and filtered (0.45 µm PTFE); diluted 1:10 in mobile phase A. Quantified against external EDQM CRS calibration; n = 3 injections.	
Chromatographic conditions: Column C ₁₈ 2.7 µm (150 × 4.6 mm) · Mobile phase 0.1 % TFA in acetonitrile / water (41:59, V/V), isocratic · Column temperature 35 °C · Flow 2.0 mL/min · Detection UV 228 nm (DAD) · Injection 5 µL · Run time 14 min.	

02 Result(s) · Quantitative Cannabinoid Assay | Резултат(и)

№	Cannabinoid · Form	RT (min)	Acceptance Criterion	Result · % w/w (mean, n = 3)	Status
4a	Δ ⁹ -THC — neutralDecarboxylated Δ ⁹ -tetrahydrocannabinol	6.42	— (component)	1.23	—
4b	Δ ⁹ -THCA-A — acidicTetrahydrocannabinolic acid A precursor	8.10	— (component)	17.90	—
4	Total Δ ⁹ -THCneutral + 0.877 × acidic	—	13.0 – 17.0 % (15.0 % ± 2.0 %)	16.93	CONFORMS
5a	CBD — neutralCannabidiol	4.05	— (component)	0.03	—
5b	CBDA — acidicCannabidiolic acid precursor	5.60	— (component)	0.05	—
5	Total CBDneutral + 0.877 × acidic	—	< 1.0 %	0.07	CONFORMS
6	CBN — CannabinolRelated substance · degradation marker	7.35	≤ 1.0 %	N.D. (< 0.05)	CONFORMS

Assay disposition — all cannabinoid results within acceptance criteria

CONFORMS

SST Replicate precision & system suitability (n = 3): Total Δ⁹-THC — T1 16.88 · T2 16.95 · T3 16.96 · mean **16.93 %** · **RSD 0.25 %** (≤ 2.0 %). Resolution Δ⁹-THC / CBD ≥ 2.0; standard similarity 100.4 % (100 ± 2.0 %); LOQ 0.05 % w/w (CBN < LOQ).

03 Method Validation · System Suitability · Measurement Uncertainty | Валидација на метод · Соодветност на системот · Мерна неодреденост

Method Status Статус на метод
Validated · in-house per ICH Q2(R2) · Validation dossier MV-PP-HPLC-001 v.03 . Specificity: Δ ⁹ -THC, CBD, CBN, THCA-A, CBDA, CBG, CBC baseline-resolved; no matrix interference. Linearity r ² ≥ 0.999 (0.05–120 % target; all five grades 9–27 %). Accuracy 98–102 % recovery. LOD 0.015 % · LOQ 0.05 % w/w.
System Suitability Соодветност на системот
HPLC-DAD (S/N HPLC-PP-02 · Cal-Cert 2026-01) · isocratic C ₁₈ 2.7 µm, 35 °C, 2.0 mL/min, 228 nm. Per session (n = 6): RSD of peak area & RT ≤ 2.0 % · resolution ≥ 2.0 · peak-to-valley ≥ 1.5 · standard similarity 100 ± 2.0 % . CRS EDQM-traceable (Δ ⁹ -THC, CBD, CBN); lot & expiry logged.
Measurement Uncertainty Мерна неодреденост
Expanded uncertainty on the assay result ± 2.0 % (k = 2, ≈ 95 % confidence). System precision RSD 0.02 % · repeatability RSD ≤ 0.5 % · intermediate precision (2 analysts, 2 days) within ± 2 %. Total Δ ⁹ -THC 16.93 % ± 0.34 % (16.6–17.3 %) evaluated against 13.0–17.0 %; reported mean conforms.

Conforms

Result: The tested sample **conforms** to the product specification **QCSP-IMB-001 v.02 §4–6 (Assay — Cannabinoid Content)** per Ph. Eur. 11.5 Monograph 3028 + general chapter 2.2.29 — Total Δ⁹-THC **16.93 %** (within 13.0–17.0 %;

label 15.0 % ± 2.0 %, **Grade IV**), Total CBD **0.07 %** (< 1.0 %), Total CBN **N.D.** (< **0.05 %**) (≤ 1.0 %). QC disposition only; final batch release is certified separately by the QP in linked CoQ CoQ-PP-2026-0005.

Notes | Забелешки

1. These results relate **only to the tested sample(s)** as received in the QC Laboratory.
2. **No part of this report may be reproduced** except in full, without prior written permission of the QC Laboratory.
3. Cannabinoid contents are reported as % w/w **on the sample as received**; totals apply the 0.877 decarboxylation factor to the acidic precursors.
4. Testing was performed **in-house** by Purely Plant QC personnel under an **ISO/IEC 17025-aligned** quality system within the **MK GMP** facility, using the in-house-validated Ph. Eur. 2.2.29 HPLC assay (dossier MV-PP-HPLC-001). **No analyses outsourced.** OOS handling per QCSOP-014.

Analysis Performed by

Анализата ја изврши

Analyst · QC Laboratory

Аналитичар · Лабораторија КК

Stojanka Pavlova

Lab ID PP-QCL-04

Date · Датум17.01.2026

Reviewed by

Прегледал

Senior Analyst · QC Laboratory

Виш аналитичар · Лабораторија КК

Marija Trajkovska

Lab ID PP-QCL-02

Date · Датум17.01.2026

Approved by

Одобрил

Head of QC Laboratory

Раководител на Лабораторија КК

Blagoj Nikolov

QC Manager · GMP-trained

Date · Датум17.01.2026

Purely Plant DOOEL · QC Analytical Laboratory · Industriska ulica 9, br. 9,
s. Kojlija 1043 · Petrovec-Skopje, North Macedonia